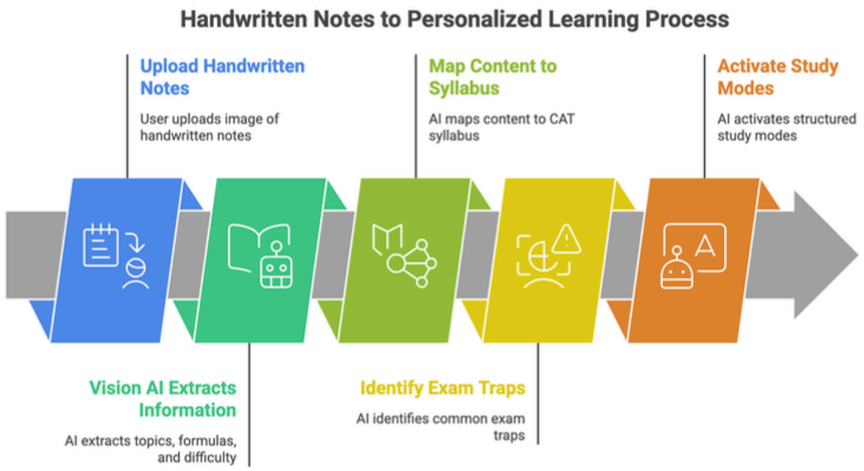
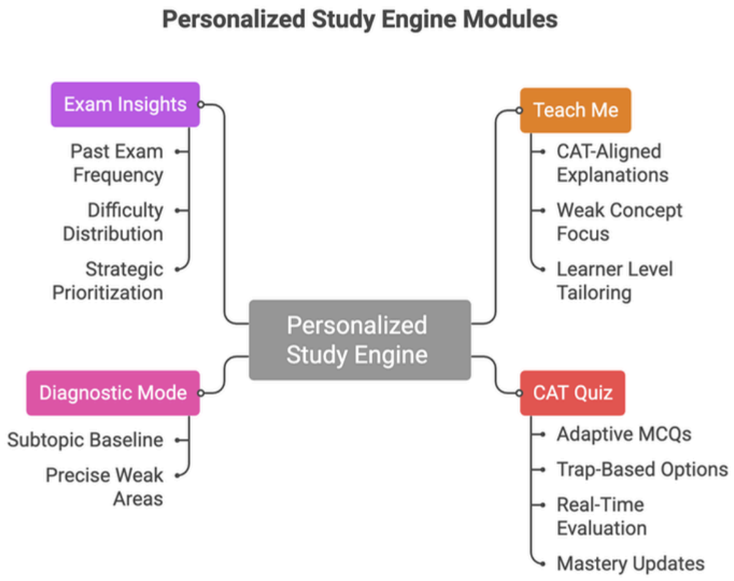
INTRODUCING CATION: YOUR PERSONALISED NOTE-MAKING STUDY PARTNER

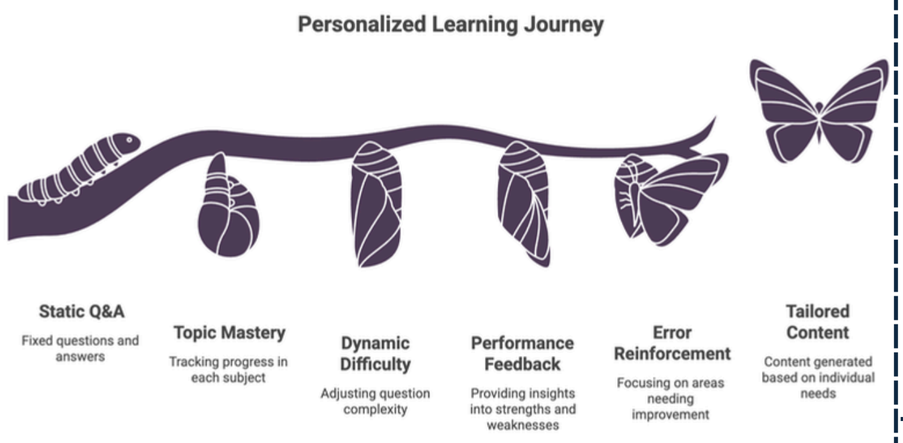
1 Core Working Flow



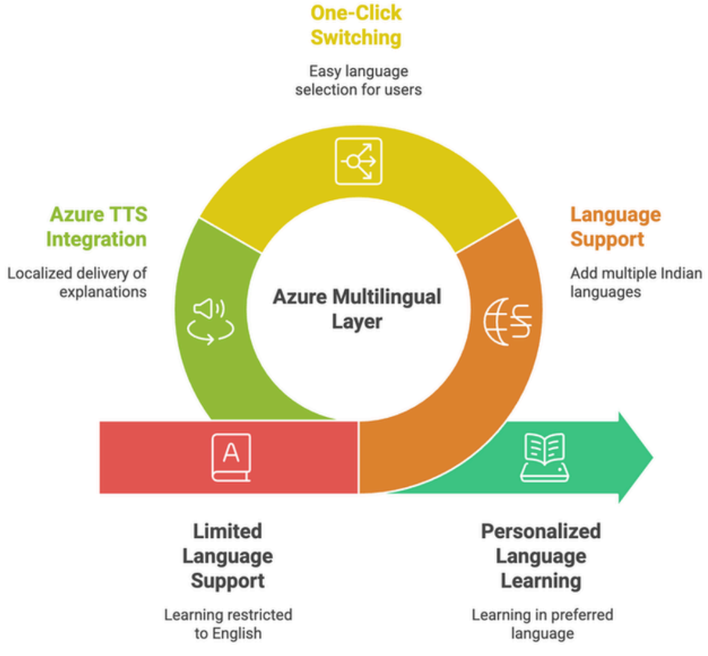
2 Personalized Study Engine



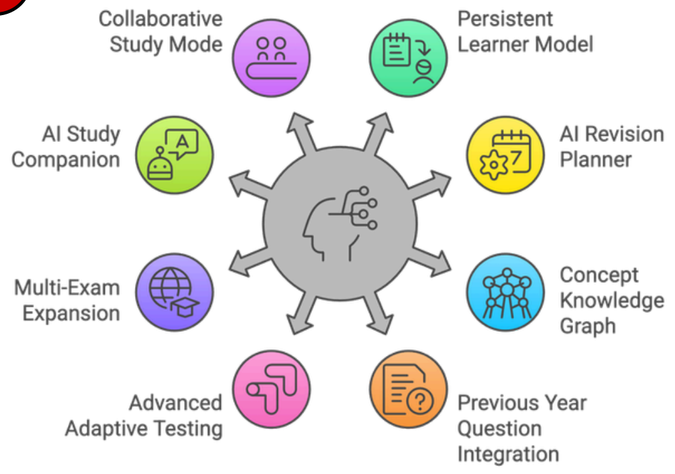
3 Personalization Layer



4 Multilingual Delivery



5 WHAT WE'RE ADDING NEXT



6 ARCHITECTURE

While the architecture is Azure-native and production-ready for Azure OpenAI, this working prototype uses **Gemini Flash 2.5** as the AI inference layer due to **student-tier Azure OpenAI deployment restrictions**. The system remains fully modular, and all orchestration, personalization, syllabus mapping, and multilingual delivery layers operate within the designed Azure stack.

GITHUB REPOSITORY LINK - CATION

Layered Architecture		
Client Interaction Layer	Mobile App (React Native) Web App (Next.js)	• Capture handwritten notes • Send questions • Play audio explanations • Display quizzes
Edge & Access Control Layer	Azure API Management Azure Active Directory B2C	• Authenticate user • Validate request • Route request to correct backend service
Application & Orchestration Layer	Azure Functions Azure Container Apps	• Orchestrate AI services • Manage workflows • Coordinate knowledge retrieval • Update student profile
AI Intelligence Layer	Azure OpenAI GPT-4o Azure OpenAI Vision	GPT-4o: • Generate explanations • Generate quizzes • Generate study plans
Knowledge & Retrieval Layer	Azure AI Search Azure OpenAI Embeddings	• Retrieve exam syllabus data • Retrieve relevant exam questions • Retrieve concept difficulty levels
Personalization & Learning Layer	Azure CosmoDB Azure ML Event Hub	Store: 1. Student scores 2. Learning History Predict: 1. Weak areas 2. Optimal revision timing Update: Student learning profile after interaction
Data Persistence Layer	Azure Blob Storage Azure SQL Database	Stores: 1. Note Images 2. Student Profiles 3. Curriculum